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Air quality prediction-based big data analytics using hebbian concordance and attention-based long short-term memory

Sathishkumar Sekar^{1,2✉} & Zhang Wei¹

With the instantaneous economic development, air quality keeps on dwindling. Some key factors for the emergence and evolution of air pollution are high-intensity pollution emissions and adverse weather circumstances. In air pollutants, Particulate Matter (PM) possessing less than 2.5 μ m is considered the most severe health issue, resulting in respiratory tract illness and cardiovascular disease. Therefore, it is mandatory to predict PM 2.5 concentrations accurately to ward off the general public from the desperate influence of air pollution in advance owing to its complex nature. Aiming at the complexity of air quality prediction, a new method called Hebbian Concordance and Attention-based Long Short-Term Memory (HC-ALSTM) is proposed. The HC-ALSTM method is split into four sections. They are preprocessing using the Statistical Normalization-based Preprocessing model, feature extraction employing the Generalised Hebbian Spatio Temporal Feature extraction model, feature selection using Concordance Correlation function, and Attention-based Long Short-Term Memory for air quality prediction. First, the Statistical Normalization-based Preprocessing model is applied to the raw dataset to normalize the impact of distinct air pollutants on the bordering factor. Second, with the Generalised Hebbian Spatio Temporal Feature extraction algorithm, processed samples are applied to extract the dimensionality-reduced spatio-temporal feature. Third, with the extracted features, essential or significant features are selected using Concordance Correlation analysis that determines the impact of pollutant concentration of bordering places for predicting air quality index involving both city and state, daily and hourly. Finally, Attention-based Long Short-Term Memory is applied to the extracted and selected features to predict air quality accurately. Through evaluation and analysis using two other evaluation methods, the proposed HC-ALSTM method performs better in error and time. Our method has dramatically improved air quality prediction accuracy and overhead compared with other methods.

Keywords Statistical normalization, Generalised hebbian, Spatio temporal, Concordance correlation, Attention mechanism, Long short-term memory

Over the past few years, swift socio-economic evolution and urbanization have brought about acute air pollution globally in distinguished cities in India, such as New Delhi. A great deal of unpleasant health consequences, like respiratory diseases, mental health issues, and mortality, were correlated with the surge in the volume of pollutant vulnerabilities. Furnishing information concerning city-wide air pollutants has paramount inferences on healthy living and the quality of life. On the one hand, provisioning air pollutant information in an accurate fashion aids children and the elderly by keeping them away from traveling to highly polluted places. On the other hand, exquisite air pollutant forecasts can ease public health policymaking by providing traffic control plans of action over highly polluted areas.

A hybrid Convolutional Neural Network-based Long Short-Term Memory (CNN-LSTM) method for providing fine-grained estimation of air pollution towards forecasting was presented in¹. Here, domain-specific features like air pollution, transport, and weather time-sensitive features were employed in addition to the

¹School of Software, East China University of Technology, No. 418, Guanglan Avenue, Economic Development District, Nanchang City, Jiangxi Province, China. ²School of Software, East China University of Technology, No.418, Guanglan Avenue, Economic Development District, Nanchang City, Jiangxi Province, China. ✉email: 201864005@ecut.edu.cn

convolution layer to improve spatial and temporal relation learning rate. Experiments revealed high accuracy in forecasting temporal and spatial features for monitoring and forecasting air pollution.

A distributed machine learning technique, spatiotemporal data partitioning, was proposed in² to address significant air quality data imbalance and spatial heterogeneity features. Here, the load balancing necessitated by distributed mechanisms were addressed by employing the Distributed Random Forest algorithm. This algorithm ensured air quality prediction via visualization functions, improving prediction accuracy execution time with considerable speedup gain.

Atmospheric chemistry mechanisms are said to be computationally extravagant and frequently depend on old-fashioned emissions information. A deep learning system that combined neural networks, fuzzy inference, and wavelet transforms was proposed in³ to predict the most distinguished air pollutant influencing Delhi, India. The chief facets of air quality mechanisms were included, which, in turn, minimized the computational time, in addition to minimum resource utilization and high spatial resolution for air quality forecasting.

The difference in Particulate Matter (PM) 2.5 depends on numerous aspects, like meteorology and other pollutant concentrations in urban areas. In⁴, a Deep learning approach was proposed to predict the hourly forecast of PM2.5 concentration in Beijing, China. Here, CNN-LSTM was employed, including a spatial-temporal feature by integrating the pollutants' historical data, meteorological data, and the PM2.5 concentration in nearby adjacent stations. As a result, accurate prediction was ensured. In⁵, the LSTM model was employed to predict PM2.5 concentrations in five representative Chinese cities.

Air quality monitoring in an accurate fashion is of high significance in our day-to-day life. Air quality prediction in advance can make timely warnings and reduce the life threat. Based on time prediction, long short-term memory (LSTM), with multi-output and multi-index of supervised learning (MMSL), was proposed in⁶. Though prevailing research works have proliferation sequence to sequence prediction for air quality, specific problems still exist. Position embedding was introduced in⁷ to assist the fully connected encoder in identifying sequential relationships between source sequences, thereby minimizing error accumulation. However, another three-dimensional air quality presentation was presented in⁸ for constructing a deep model employing a Spatio-temporal collaborative technique for predicting regional air quality. The advantage of this method lies in dealing with multiple air quality features.

The ever-increasing city population and the elevated motorization level contribute to the ever-increasing traffic volume and, therefore, the ever-increasing emission of gases, causing hazardous effects. In⁹, a time series forecasting model based on multi-criteria for air quality prediction was proposed, ensuring reliable prediction using spatial and temporal relationships. However, another paper integrating deep learning techniques and feature selection based on statistical approaches was presented in¹⁰. In¹¹, the deep learning technique was presented to predict performance metrics such as delicate particulate matter and carbon monoxide. Moreover, the LSTM neural network-based model here also processed the sequences in both the forward and backward direction, ensuring timesteps in both directions.

Over the past few years, the general public has started to focus more attention on the influence of the environment on health; therefore, information about air quality has become the concentration of people's day-to-day lives. The prevailing air quality monitoring devices can provide real-time air quality data; however, they are not found to be sufficient. In recent years, high reliability and accuracy have to be ensured.

In¹², the issue of processing high-dimensional large-scale data using the Light GBM model and including forecasting data for predicting air quality was presented. By exploring forecasting data, the prediction accuracy was improved by fully utilizing spatial data.

In the face of the above-said issues, in this work, a novel method called HC-ALSTM is proposed to predict the PM 2.5 concentrations accurately. In dealing with the normalization of raw data, we propose a Statistically normalized preprocessing model to construct high-dimensional time features. In the Spatio-temporal dimension, we model each air quality monitoring station and utilize the relevant information extracted from the Generalised Hebbian function. Finally, with the significant selected feature, air quality prediction is performed using Attention-based Long Short-Term Memory.

The main contributions of this article include:

- We propose HC-ALSTM method of constructing features and extend to develop more data samples for air quality prediction. The correlation of PM2.5 concentration in time series is explored to normalize its trend by employing Statistical Normalization-based Preprocessing.
- We select the Generalised Hebbian Spatio Temporal dimension reduction model to extract the Spatio-temporal features in the weather forecast information to minimize the negative influence of redundant and irrelevant features. In the experiment, it is verified that the spatiotemporal dimension can improve the prediction effect.
- According to the contribution of the feature group using Concordance Correlation analysis, we select the specific features according to the feature's correlation levels to minimize the training time involved.
- Finally, we propose Attention-based Long Short-Term Memory for air quality prediction, in which the context vector obtains the aggregate sum of hidden states to predict air quality accurately.
- The experiment shows that the multi-model fusion scheme has a better effect on multiple indicators than a single model.

The remainder of this paper is organized as follows. Section "Related works" describes related works, and Sect. "Methodology" defines the problem to be addressed along with the problem statement and motivation. Moreover, an overview of the proposed method, as well as the diagram and algorithm, is presented. Section "Experimental settings" provides the experimental setup and compares the proposed method's performance with

previous traditional and state-of-the-art methods based on real-world datasets in Sect. "Discussion". Finally, Sect. "Conclusion" summarizes and concludes the paper.

Related works

However, another attention-based air quality predictor was presented in¹³ to focus on integrating several neural networks comprising an artificial neural network, a convolutional neural network, and a long-short-term memory to extract spatial-temporal relations, ensuring excellent performance. In¹⁴, a deep learning hybrid CLSTM model was proposed. Here, CNN was integrated with the LSTM network to forecast hourly.

Significant air quality predictions over the past few years have become one of the hot research topics. However, prediction remains a major issue due to data source instability and the differences between pollutant concentrations. In¹⁵, a case study of air quality prediction in Indian cities was investigated. In¹⁶, a hybrid method of Vector Auto Regression (VAR) and LSTM method was presented to enhance the Air Quality Index performance.

With the evolution of data-driven modelling mechanisms, utilizing the neural network to enhance the transport procedure of atmospheric pollutants and constructing PM_{2.5} for time series prediction over the past few years have become major research topics. In¹⁷, a LSTM with an encoder multitask learning mechanism was presented to predict PM_{2.5} time series in several locations. However, another integrated model was proposed in¹⁸ using attention-based deep learning for accurate air quality prediction.

Over the past few years, conditions related to meteorological factors have significantly impacted air quality and, hence, play a significant role in air quality prediction. However, due to the deficient black box nature, trustworthiness is hard to attain. The impact of meteorological conditions on air quality prediction was revealed in¹⁹ by employing explainable deep learning to address this aspect. With this prediction, the accuracy of air quality was said to be improved. The chi-square test was introduced in²⁰ for air quality prediction in the developing industry. Air quality prediction using stacked ResNet and coral alignment was utilized in²¹ in case of non-linear correlation between the particulate matter.

A hybrid intelligent method based on LSTM and multi-verse optimization algorithm (MVO) was developed in²² to predict and analyze air pollution acquired from Combined Cycle Power Plants. With this, accuracy with minimum error was said to be ensured. Deep convolutional networks were applied in²³ for large-scale areas spread sparsely. An ensemble method was applied in²⁴ using the decomposition and reconstruction technique to concentrate on both the error and accuracy aspects. However, another LSTM work was investigated in²⁵ for air quality prediction.

Deep learning approaches were introduced in²⁶ to enhance the air quality prediction performance. Air pollutant data were investigated in²⁷ through different waves. Study considers short-term forecasting of cumulative reported incidences and mortality in²⁸. Multivariate time series short-term forecasting was analyzed with less time. Empirical Mode Decomposition (EMD), Transformer, and Bidirectional Long Short-Term Memory Neural Network (BiLSTM) were discussed in²⁹ for achieving the predicted AQI values. But, the time was not minimized.

The air quality time series prediction fusion model was introduced in³⁰ with the D-S evidence theory. A hybrid deep learning model was integrated into³¹ with variational mode decomposition (VMD), graph attention networks (GAT), and BiLSTM named VMD-GAT-BiLSTM. VMD was employed to decompose the original PM_{2.5} data. GAT was utilized to find the deep spatial relations between dissimilar monitoring stations. BiLSTM was applied to select the temporal features and obtain air quality prediction results. However, the computational overhead was not reduced.

Motivated by the materials and methods mentioned above, this work proposes big data analytics using air data for accurate and precise air quality prediction. An elaborate description is provided in the following sections.

Methodology

This work presents a coherent multi-model method to improve air quality prediction accuracy. In this section, the detailed methodology of the proposed work is narrated. The motivation and problem statement are initially explained, followed by other sections. In the following subsections, an outline of the proposed work is explored; in the preceding subsections, the normalization issue is resolved; following, associated with the work, the association analysis is brought off, the impact of the bordering stations is assessed, and finally, the prediction model is illustrated elaborately.

Problem statement and motivation

Predicting air pollution has been a demanding task over the past few years owing to numerous factors like an exponential increase in population, large-scale construction activities, land-locked geography, and so on. Air pollution forecasting accelerates perpetually environmental vibrations and ensures better health for the general public. Samples concerning pollutants and meteorological factors are obtained daily and hourly to forecast air pollution. Therefore, the pollution-related dataset is said to be a multifaceted time-series data. The existing methods have addressed air quality prediction in a significant manner. However, some issues still have to be solved. The existing methods focused on the air quality index values by considering the pollutant concentration and meteorological factors of that specific city or station. However, it remains a significant issue to address the bordering impact while predicting the air quality index of that specific city or station. Another challenge is addressing the error during air quality prediction involving time-series data. The main objective of this work is to develop a robust deep learning-based forecasting method that can predict a place's air quality in a more accurate and timely fashion by considering the pollutant concentrations of the bordering cities or stations.

Overview of the proposed work

This study proposes a Hebbian Concordance and Attention-based Long Short-Term Memory (HC-ALSTM).

Figure 1 shows the architecture diagram of the HC-ALSTM method. It consists of four phases: sample data normalization, spatiotemporal feature extraction, correlation analysis-based feature selection, and deep learning model construction. Various influencing factors, such as city, station, and boundary area, are utilized as input. In the sample data normalization phase, a Statistical Normalization-based Preprocessing algorithm is proposed to obtain stable processed output data with minimum error. In the next phase, the Spatio-temporal data directly or indirectly influencing air quality are extracted using the Generalised Hebbian Spatio Temporal Feature extraction model. The third part identifies the influence of correlation between distinct relevant and significant particles with meteorological data of bordering places for predicting air quality index, which is selected without overlooking its influences on AQI evaluation. Finally, Attention-based Long Short-Term Memory integrates the prediction results. The best prediction results obtained via the attention mechanism are used as the final prediction value of air quality data.

Statistical normalization-based preprocessing model

To demonstrate data preprocessing, we used the Air Quality Data in India dataset. Air Quality Index (AQI) and hourly data across stations and cities in India employed from 2015 to 2020. The cities included are, Ahmedabad, Aizawl, Amaravati, Amritsar, Bengaluru, Bhopal, Brajrajnagar, Chandigarh, Chennai, Coimbatore, Delhi, Ernakulam, Gurugram, Guwahati, Hyderabad, Jaipur, Jorapokhar, Kochi, Kolkata, Lucknow, Mumbai, Patna, Shillong, Talcher, Thiruvananthapuram, Visakhapatnam. Both the daily and hourly average concentrations of delicate particulate matter (PM_{2.5}), inhalable particulate matter (PM₁₀), sulfur dioxide (SO₂), nitrogen dioxide (NO₂), ozone (O₃), carbon monoxide (CO), etc., selected from 2015 to 2020 are used as an original dataset, which is denoted by 'DS'. Several parts of the dataset 'DS' are shown in Table 1.

In our work, the Global Air Pollution Dataset is used and it is considered from the <https://www.kaggle.com/datasets/hasibalmuzdadid/global-air-pollution-dataset>. The dataset includes 23,463 records with 12 attributes. Table 1 presents the attributes of the data set used to predict air pollution.

As given in the above table, 16 features ' $F = \{F_1, F_2, \dots, F_m\}$ ' or attributes '*i.e.*, $m = 16$ ' are present. Using these 16 features, air quality prediction is made. Moreover, Table 2 given below lists the Air Quality Data in India dataset station file. Here, the corresponding five features ' $SF = \{SF_1, SF_2, \dots, SF_{sm}\}$ ' or attributes '*i.e.*, $sm = 5$ ' are present. Accordingly, the station or city's daily and hourly recordings are maintained and mentioned in Table 3.

With the above details, the corresponding four input data matrices, *i.e.*, city_day 'CD', city_hour 'CH', station_day 'SD', and station_hour 'SH' are formulated separately.

$$CD = \begin{bmatrix} CD_1F_1 & CD_1F_2 & \dots & CD_1F_m \\ CD_2F_1 & CD_2F_2 & \dots & CD_2F_m \\ \dots & \dots & \dots & \dots \\ CD_{cds}F_1 & CD_{cds}F_2 & \dots & CD_{cds}F_n \end{bmatrix} \quad (1)$$

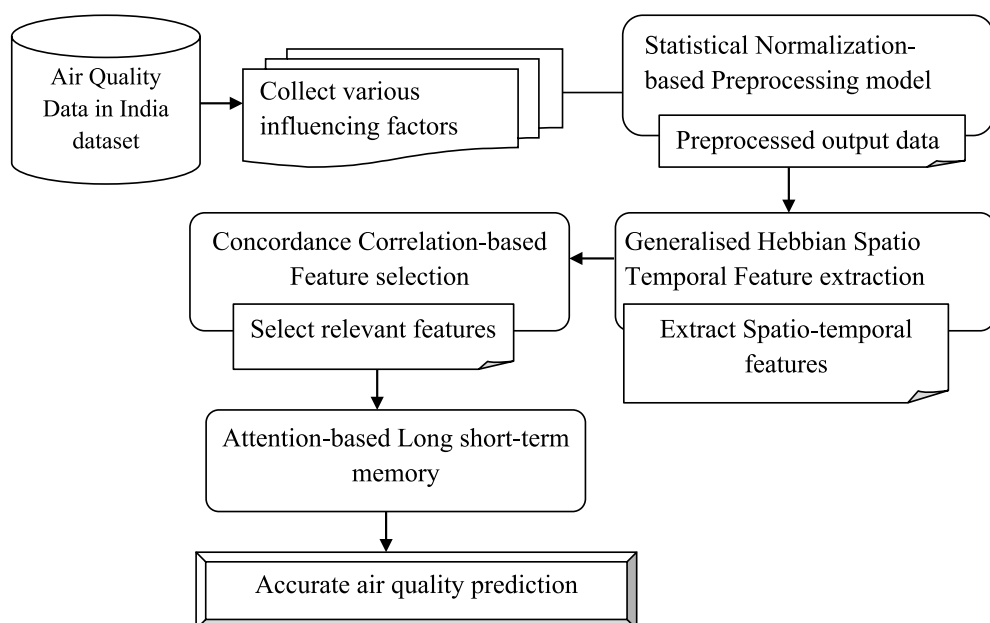


Fig. 1. Architecture of the proposed HC-ALSTM method.

S. No	Feature or attributes	Description
1	City	Name of city
2	Date	Date gathered
3	PM2.5	Particulate matter 2.5 μm in ug
4	PM10	Particulate matter 10 μm in ug
5	NO	Nitric oxide in ug
6	NO2	Nitric dioxide in ug
7	NOx	Any nitric x-oxide
8	NH3	Ammonia in ug
9	CO	Carbon monoxide in ug
10	SO2	Sulphur dioxide in ug
11	O3	Trioxxygen in ug
12	Benzene	Benzene
13	Toluene	Toluene
14	Xylene	Xylene
15	AQI	Air quality index
16	AQI_Bucket	AQI category [good, satisfactory, moderately polluted, poor, very poor, severe]

Table 1. Air quality data in India dataset description.

S. No	Features	Description
1	Country	Name of the country
2	City	Name of the city
3	AQI Value	Overall AQI value of the city
4	AQI Category	Overall AQI category of the city
5	CO AQI Value	AQI value of Carbon Monoxide in the city
6	CO AQI Category	QI category of Carbon Monoxide of the city
7	Ozone AQI Value	AQI category of Ozone of the city
8	Ozone AQI Category	AQI category of Ozone of the city
9	NO2 AQI Value	AQI value of Nitrogen Dioxide in the city
10	NO2 AQI Category	AQI category of Nitrogen Dioxide of the city
11	PM2.5 AQI Value	QI value of Particulate Matter with a diameter of 2.5 μm or less of the city
12	PM2.5 AQI Category	QI category of Particulate Matter with a diameter of 2.5 μm or less of the city

Table 2. Attributes information of the global air pollution dataset.

S. No	Feature or attributes	Description
1	Station_ID	Station identification
2	Station_Name	Name of station
3	City	Name of city
4	State	Name of state
5	Status	Status of the corresponding station

Table 3. Description of the station [air quality data in India dataset].

$$CH = \begin{bmatrix} CH_1F_1 & CH_1F_2 & \dots & CH_1F_m \\ CH_2F_1 & CH_2F_2 & \dots & CH_2F_m \\ \vdots & \vdots & \ddots & \vdots \\ CH_{chs}F_1 & CH_{chs}F_2 & \dots & CH_{chs}F_n \end{bmatrix} \tag{2}$$

$$SD = \begin{bmatrix} SD_1F_1 & SD_1F_2 & \dots & SD_1F_m \\ SD_2F_1 & SD_2F_2 & \dots & SD_2F_m \\ \vdots & \vdots & \ddots & \vdots \\ SD_{sds}F_1 & SD_{sds}F_2 & \dots & SD_{sds}F_n \end{bmatrix} \tag{3}$$

$$SH = \begin{bmatrix} SH_1F_1 & SH_1F_2 & \dots & SH_1F_m \\ SH_2F_1 & SH_2F_2 & \dots & SH_2F_m \\ \vdots & \vdots & \ddots & \vdots \\ SH_{shs}F_1 & SH_{shs}F_2 & \dots & SH_{shs}F_n \end{bmatrix} \quad (4)$$

From the above formulations (1), (2), (3), and (4) four input matrices, i.e., city data recordings ‘*CD*’, city hour recordings ‘*CH*’, station day recordings ‘*SD*’ and station hour recordings ‘*SH*’ are first initialized from the Air Quality Data in India Dataset as input. Moreover, ‘*cds*’, ‘*chs*’, ‘*sds*’ and ‘*shs*’ represent the corresponding city data recording samples, city hour recording samples, station day recording samples and station hour recording samples considered for simulation respectively. Exiting Min–max scaling was employed [32] to perform data normalization. However, the Min–max scaling cannot capture the unique characteristics and relationships between different pollutants. This can lead to challenges in predicting or modelling air quality, particularly when dealing with multiple pollutant types simultaneously. Compared to min–max normalization, Statistical normalization is the preprocessing technique to decrease RMSE for enhancing the performance of ML algorithms. Normalization helps reduce the impact of pollutant types to provide better prediction accuracy. Statistical Normalization method is chosen for the above four input matrices that assume normally distributed data and stabilize the impact of numerous air pollutant types. Figure 2 shows the structure of the Statistical Normalization-based Preprocessing model.

Structure of Statistical Normalization-based Preprocessing Model is illustrated in Fig. 2. The Air Quality Data in India dataset provided as input, the original dataset ‘*DS*’ is split into a training set ‘*DS_{train}*’ and a validation set ‘*DS_{val}*’, and a testing set ‘*DS_{test}*’ that is based on a definite percentage. The training set (70%), testing test (20%), and validation set (10%) were used. With these three sets, the normalization function stabilizes the impact of numerous air pollutant types. With the training set ‘*DS_{train}*’ acquired as sample simulation, the proposed normalization method is given below.

$$S_{train}^i = \frac{S_{train}^i - \mu_{train}}{\sigma_{train}} \quad (5)$$

From the above Eq. (5), the normalized sample data ‘*S_{train}ⁱ*’ from ‘*DS_{train}*’ is obtained based on the sample data from the training. ‘*S_{train}ⁱ*’ with ‘*i* ∈ (1, 2, ..., *n*)’, mean ‘*μ_{train}*’ of training set ‘*DS_{train}*’ and standard deviation ‘*σ_{train}*’ of training set ‘*DS_{train}*’, respectively. The pseudo-code representation of statistical normalization-based preprocessing is illustrated below.

Input: Dataset ‘ <i>DS</i> ’, Features ‘ <i>F</i> = { <i>F</i> ₁ , <i>F</i> ₂ , ..., <i>F</i> _{<i>m</i>} }’, Station Features ‘ <i>SF</i> = { <i>SF</i> ₁ , <i>SF</i> ₂ , ..., <i>SF</i> _{<i>sm</i>} }’, Samples ‘ <i>S</i> = { <i>S</i> ₁ , <i>S</i> ₂ , ..., <i>S</i> _{<i>n</i>} }’
Output: Balanced equilibrium-based preprocessed sample results ‘ <i>PS</i> ’
1: Initialize ‘ <i>m</i> = 16’, ‘ <i>sm</i> = 5’, city day sample ‘ <i>cds</i> ’, city hour sample ‘ <i>chs</i> ’, stations day sample ‘ <i>sds</i> ’, station hour sample ‘ <i>shs</i> ’, ‘ <i>n</i> ’ 2: Initialize training set ‘ <i>DS_{train}</i> ’, validation set ‘ <i>DS_{val}</i> ’, testing set ‘ <i>DS_{test}</i> ’ 3: Begin 4: For each Dataset ‘ <i>DS</i> ’ with Features ‘ <i>F</i> ’ and Station Features ‘ <i>SF</i> ’ 5: Formulate city_day ‘ <i>CD</i> ’ as given in equations (1) 6: Formulate city_hour ‘ <i>CH</i> ’ as given in equations (2) 7: Formulate station_day ‘ <i>SD</i> ’ as given in equations (3) 8: Formulate station_hour ‘ <i>SH</i> ’ as given in equations (4) 6: Apply statistical normalization ‘ <i>S_{train}ⁱ</i> ’ as given in equation (5) 7: Return preprocessed or processed sample results ‘ <i>PS</i> ’ 8: End for 9: End

Algorithm 1. Statistical Normalization-based Preprocessing Model.

Step-by-step process of 1 Statistical Normalization-based Preprocessing Model is illustrated in algorithm 1. As given in the above algorithm 1, four input vectors are initially formulated from the raw dataset to reduce the RMSE and balance the impact of different pollutants during air pollution forecasting. Next, statistical-based normalization is applied with the obtained input vector matrices so that stable processed output with minimum error factor is generated.

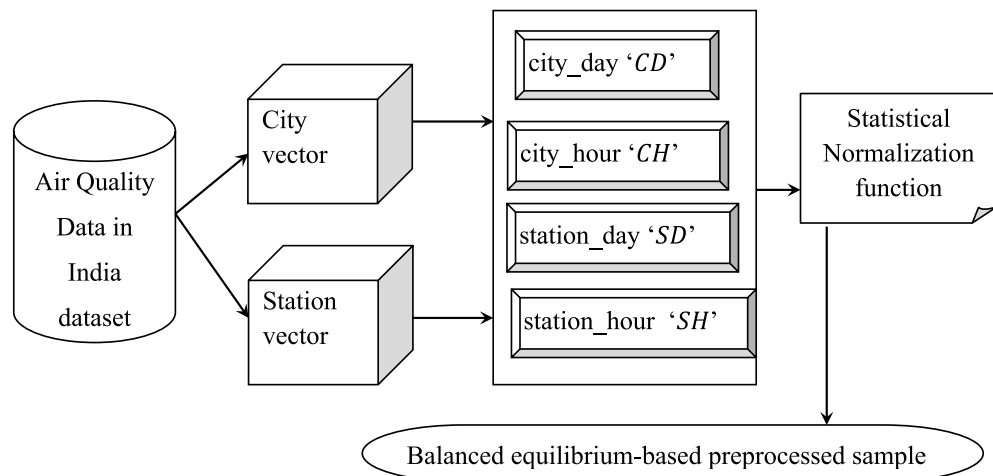


Fig. 2. Structure of Statistical Normalization-based Preprocessing Model.

Generalised hebbian spatio temporal feature extraction

The spatial data, similar to boundary area, directly or indirectly influence air quality, so the proposed method utilizes these spatial data as supplementary information for air quality prediction. LSTM focuses on learning temporal patterns in time series data, such as air pollution levels at specific time intervals. However, LSTM cannot capture the spatial relationships. They can learn correlations between dissimilar locations over time, but they don't inherently model spatial dependencies like a boundary area's influence. Also, these spatial data cannot be provided as input directly into the LSTM. Contrary to existing LSTM, Generalised Hebbian Spatio-temporal feature extraction is employed to provide air quality prediction by considering Spatio-temporal relationships with boundary areas. Also, the Generalised Hebbian function is used to embed the information composed of the coordinates, boundary area, and station name of the prediction area to enhance the prediction effect for air quality. This approach helps to more accurate air quality predictions. Figure 3 shows the information diagram of spatial information such as station class, boundary class, and prediction area.

An information diagram between the station class, boundary class, and prediction area is provided in the above Fig. 3. Graph ' $G_{sb}(s \cup s, \beta_{sb})$ ' between prediction station and boundary area denotes the distance relationship of each station concerning its boundary area, where ' s ' represents the station to be predicted, ' e_{ij} ' represents the edges between any two stations, boundary area, and weight ' w_{ij} ' denotes the distance between the station and the boundary area. Second, the graph ' $G_{bc}(b \cup c, \beta_{bc})$ ' between the boundary and city denotes the distribution of boundary areas in the prediction city, where ' b ' denotes the classes of boundary areas, and the classes ' $b_1, b_2, \dots, b_n$ ' respectively. Here, ' β_{bc} ' denotes the set of edges ' e_{ij} ' between the boundary class and city, and its weight ' w_{ij} ' denotes the number of boundary areas comprising the class ' b_i ' in the prediction area. Finally, the graph ' $G_{cs}(s \cup sg, \beta_{cs})$ ' between the city and station that associates the city and station names in

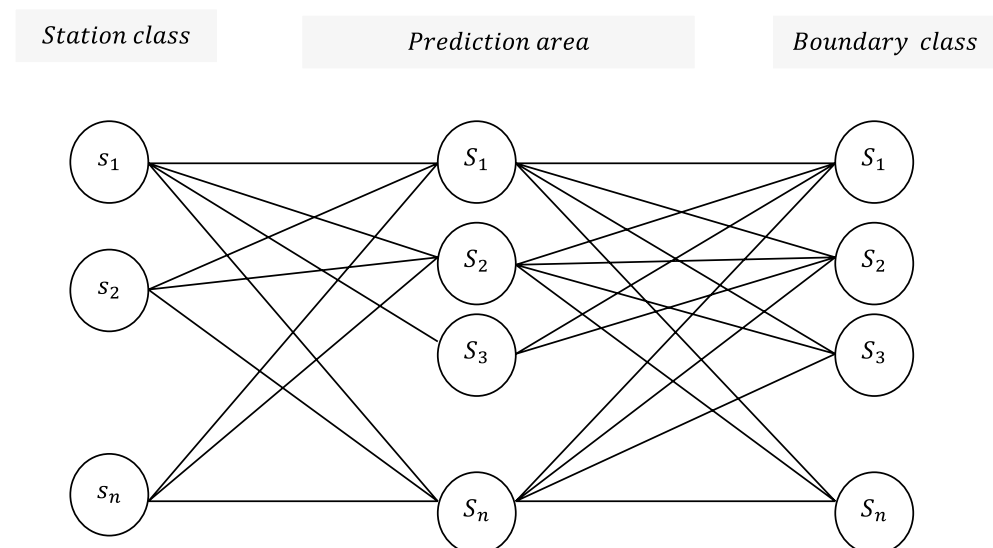


Fig. 3. Information diagram between the station class, boundary class, and prediction area.

the prediction area, where 's' denotes the set of station classes, ' β_{cs} ' denotes the set of edges ' e_{ij} ' between the city and station-classes and its weight ' w_{ij} ' denotes the total sample size 'n' included in the prediction area.

$$L(G_{sb}) = - \sum_{e_{ij} \in \beta_{sb}} w_{ij} \log p(v_j | v_i) \quad (6)$$

$$L(G_{bc}) = - \sum_{e_{ij} \in \beta_{bc}} w_{ij} \log p(v_j | v_i) \quad (7)$$

$$L(G_{cs}) = - \sum_{e_{ij} \in \beta_{cs}} w_{ij} \log p(v_j | v_i) \quad (8)$$

$$L = L(G_{sb}) + L(G_{bc}) + L(G_{cs}) \quad (9)$$

From the above Eqs. (6), (7), (8), and (9), the low dimension vector representation between stations and boundary area, ' $L(G_{sb})$ ', low dimension vector representation between boundary and city, ' $L(G_{bc})$ ' and finally, low dimension vector representation between city and station ' $L(G_{cs})$ ' are obtained. These three vectors aggregated value ' L ' are finally provided as input to model the Generalised Hebbian function. The Generalised Hebbian function is formulated as given below.

$$y(L) = \sum_{i,j=1}^{m,n} L_{ij} w_{ij} \quad (10)$$

$$\frac{dw}{dt} = \eta t(t) [L(t) - y(t) w(t)] \quad (11)$$

From the above Eqs. (10) and (11), the aggregated results ' $L(t)$ ' at a time ' t ' based on the synaptic weight ' w_{ij} ' are provided as input for the Generalised Hebbian function. The derivation of weight is formulated by employing the learning rate ' η ' to extract both time ' t ' and non-time ' $L(t)$ ' (i.e., the boundary of city and station) series-based essential features ensure accurate air quality prediction. Based on the synaptic strengths ' w_{ij} ' in the brain that is updated according to the experience, i.e., that changes are proportional to the correlation between the city, station, and boundary. The pseudo-code representation of Generalised Hebbian Spatio Temporal Feature extraction is described below.

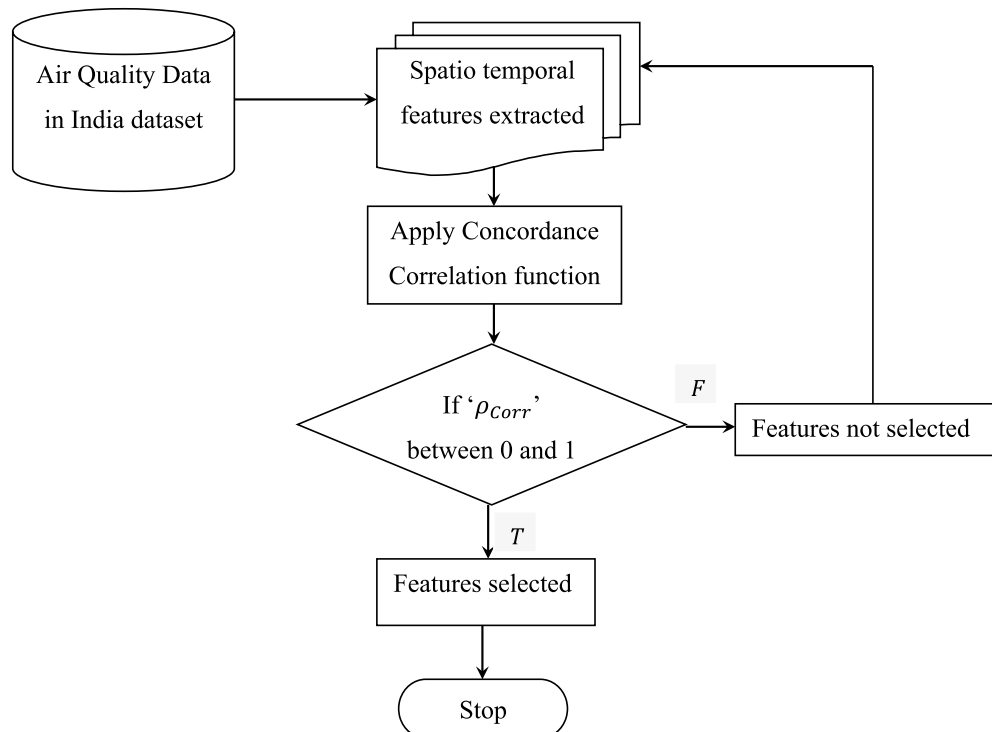


Fig. 4. Structure of Concordance Correlation-based Feature Selection.

Input: Dataset ' DS ', Features ' $F = \{F_1, F_2, \dots, F_m\}$ ', Station Features ' $SF = \{SF_1, SF_2, \dots, SF_{sm}\}$ ', Samples ' $S = \{S_1, S_2, \dots, S_n\}$ ', Balanced equilibrium-based preprocessed sample results ' PS '
Output: Dimensionality-reduced spatiotemporal feature extracted ' FE '
1: Initialize station ' s ', city ' c ', boundary or boundary area ' b ', learning rate ' $\eta = 0.01$ ', time ' t ' 2: Begin 3: For each Dataset ' DS ' with Features ' F ', Station Features ' SF ' and processed sample results ' PS ' 4: Evaluate low-dimension vector representation between stations and boundary area ' $L(G_{sb})$ ' as given in equation (6) 5: Evaluate low-dimension vector representation between boundary and city ' $L(G_{bc})$ ' as given in equation (7) 6: Evaluate low-dimension vector representation between city and station ' $L(G_{cs})$ ' as given in equation (8) 8. Formulate three vectors aggregated value ' L ' as given in equation (9) 9: Obtain aggregated vector representation 10: For each aggregated vector representation 11: Evaluate the Generalised Hebbian function ' $y(L)$ ' and ' $\frac{dw}{dt}$ ' as given in equations (10) and (11) 12: Return dimensionality-reduced features extracted ' FE ' 13: End for 14: End for 15: End

Algorithm 2. Generalised Hebbian Spatio Temporal Feature Extraction

Step-by-step process of Generalised Hebbian Spatio Temporal Feature extraction is provided in algorithm 2. As given in the above algorithm, to extract the original or actual features and minimise the dimensions of the input variables; initially, three different vector representations are formed separately based on temporal factors. The first vector is formed between stations and boundary area, the second vector is between boundary and city, and finally, the third vector is between city and station. Next, the aggregated vector representation is formulated. Finally, these aggregated vector representation forms as input to the Generalised Hebbian function to consider the spatial factors also.

Concordance correlation-based feature selection model

From the view of feature selection, distinct features have different impacts or consequences. By synthesizing the dominant feature with each supplementary feature in an equidistant fashion, the significance of the dominant feature can be obtained and encapsulate the impact of supplementary features. Inferences exhibited that the particles ' $PM_{2.5}$ ' and ' PM_{10} ' can be influenced by several quantifiable features. Nevertheless, the demanding issue remains in isolating the feature that is not useful or valuable for air quality prediction, as irrelevant features are laborious for the training phase. Hence, it becomes paramount to identify the correlation between distinct relevant and significant particles with meteorological data. Our work applies the Concordance Correlation-based Feature selection model to estimate the agreement between two variables or features with better reproducibility. Here, reproducibility refers to the results obtained via statistical analysis, which should be attained with high reliability when the observation or experiment is repeated or replicated.

A key advantage of a concordance correlation-based feature selection model is applied to evaluate both precision and accuracy. However, many other correlation-based methods may struggle to effectively capture relevant features. Contrary to existing methods, the advantage of concordance correlation-based feature selection is employed to identify relevant and eliminate redundant features. By reducing the number of features, the concordance correlation function becomes less complex and easier to understand. By choosing the subset of features that are positively correlated, the model can focus on the most important information and improve its predictive power. By using fewer features, the model requires less computational time to train and predict. Figure 4 shows the structure of the Concordance Correlation-based Feature selection model.

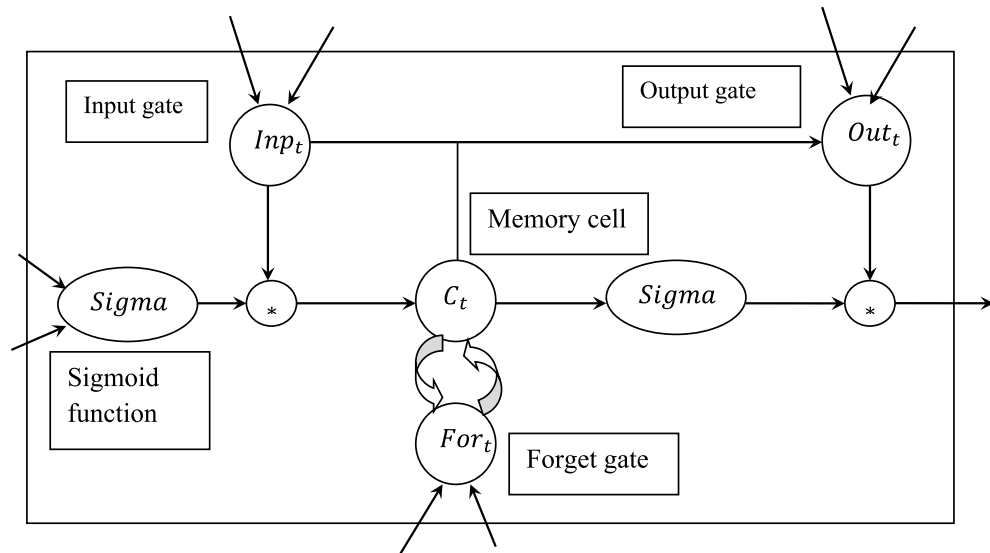


Fig. 5. Block diagram of Attention-based Long short-term memory for air quality prediction model.

Concordance Correlation-based Feature Selection structure is shown in the above Fig. 4. The Spatio-temporal extracted features provided as input, the objective of designing the Concordance Correlation-based Feature selection model remains in selecting computationally efficient features. With this objective initially, the Concordance Correlation function is applied to the extracted features. The conditionality checking follows this is made to select the relevant features or discard the irrelevant ones. Initially, the Concordance Correlation function is mathematically formulated as given below.

$$\rho_{Corr} = FE \left[\frac{2Cov_{pq}}{Var_p^2 + Var_q^2 + (p' - q')^2} \right] \quad (12)$$

From the above Eq. (12), ' p' ' and ' q' ' represents the means for the two features ' p ', ' q ' (i.e., ' $p, q \in F$ '), ' Var_p^2 ', ' Var_q^2 ' represents the corresponding variances for the two features ' p ', ' q ' and finally, ' ρ_{Corr} ' forming the correlation between two features ' p ', ' q ' respectively. The mean for the two features ' p ', ' q ' is formulated below.

$$p' = \frac{1}{N} \sum_{n=1}^N FE[p_n]; q' = \frac{1}{N} \sum_{n=1}^N FE[q_n] \quad (13)$$

With the obtained mean values from the above Eq. (13), the variances for the two features ' p ', ' q ' are mathematically evaluated below.

$$Var_p^2 = \frac{1}{N} \sum_{n=1}^N FE[(p_n - p')^2]; Var_q^2 = \frac{1}{N} \sum_{n=1}^N FE[(q_n - q')^2] \quad (14)$$

Finally, the covariance for the two features ' p ', ' q ' is obtained below.

$$Cov_{pq} = \frac{1}{N} \sum_{n=1}^N (p_n - p)(q_n - q) \quad (15)$$

With the above formulations, the prior formulation presents a positive correlation when ' $0 < \rho_{Corr} < 1$ ' and a negative correlation when ' $-1 < \rho_{Corr} < 0$ '. The pseudo-code representation of Concordance Correlation-based Feature selection is given below.

S.No	AQIValue	AirQualityLevel
1	0 – 50	Excellent
2	51 – 100	Good
3	101 – 150	Light Pollution
4	151 – 200	Moderate Pollution
5	201 – 300	Heavy Pollution
6	> 300	Serious Pollution

Table 4. AQI value and air quality level.

Software requirements	Hardware requirements
O/S: Windows 10 and above	CPU Processor: Intel® Core™ i3 Processor
Language: Python	Hard drive capacity: 1 TB
	Mouse: Logitech
	Keyboard: 110 keys enhanced
	RAM (Memory capacity): 8 GB
	System type: 32-bit Operating system

Table 5. Hardware and Software Configurations of proposed HC-ALSTM method and existing CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM.

Input: Dataset ‘ DS ’, Features ‘ $F = \{F_1, F_2, \dots, F_m\}$ ’, Station Features ‘ $SF = \{SF_1, SF_2, \dots, SF_{sm}\}$ ’, Samples ‘ $S = \{S_1, S_2, \dots, S_n\}$ ’
Output: Computationally efficient feature selection
1: Initialize dimensionality-reduced features extracted ‘ FE ’ 2: Begin 3: ForEach Dataset ‘ DS ’ with Features ‘ F ’, Station Features ‘ SF ’, features extracted ‘ FE ’ and processed sample results ‘ PS ’ 4: For each feature, ‘ p ’, ‘ q ’ 5: Evaluate the Concordance Correlation ‘ ρ_{Corr} ’ as given in equation (12) 6: If ‘ ρ_{Corr} ’ lies between 0 and 1 7: Then positive correlation is said to exist, said to be relevant features, and hence selected 8: Return features selected ‘ FS ’ 9: End if 10: If ‘ ρ_{Corr} ’ lies between -1 and 0 11: Then negative correlation is said to exist, said to be less relevant features and hence not selected 12: Return features not selected ‘ FNS ’ 13: End if 14: End for 15: End for 16: End

Algorithm 3. Concordance Correlation-based Feature selection [PM2.5, PM10, So2, Nox, Co, O3].

Step-by-step process of Concordance Correlation-based Feature selection is explained in algorithm 3. As given in the above algorithm, to select relevant features utilized for further processing, therefore reducing the

S. No	Hyperparameters	Description
1	Number of gates used	Forget gate, input gate, and output gate
2	Activation function used in forget gate	Sigmoid function
3	Activation function used in the output gate	Context vector
4	Learning rate	The value of the learning rate used in our work is 0.01
5	Number of epochs	The number of epochs is used in 20
6	Batch size	The value of the batch size used in our work is 16
7	Filter size	The size of the filter is 3×3 used in our work

Table 6. Hyper-parameters and description.

Air quality samples	Computational overhead (KB)					
	HC-ALSTM	CNN-LSTM	Spatiotemporal data partitioning	Deep learning system	Deep learning approach	EMD-Transformer-BiLSTM
5000	2500	4000	6500	8000	9500	3000
10,000	3100	4700	6800	8500	10,250	3700
15,000	3500	5200	7100	8800	10,800	4200
20,000	4100	5800	7300	9200	11,350	4800
25,000	4800	6200	7800	9500	11,500	5300
30,000	5300	6800	8100	9900	11,850	6000
35,000	5900	7300	8500	10,350	12,350	6500
40,000	6200	7700	9000	11,250	14,550	6900
45,000	6800	8100	9200	13,000	15,350	7300
50,000	7300	8800	9500	14,500	17,200	8000

Table 7. Computational overhead estimates of air quality big data analysis using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM.

training time involved and be further optimized to perform better, dynamic correlation of both ‘ $PM_{2.5}$ ’ and ‘ PM_{10} ’ particles between previous and future time points are initially captured according to the extracted feature. Next, the Concordance Correlation function is applied to the extracted feature that, with the aid of variance and covariance measure, obtains the relevant features in a timely fashion, therefore not only reducing the average training time for air quality prediction but also making the proposed method more efficient effective and less time-consuming.

Attention-based long short-term memory for air quality prediction

Distributed random forest-based approaches do poorly in predictive tasks because they do not consider the long-term dependence on spatiotemporal data and remember extended input sources. Therefore, in the context of air quality prediction, LSTM networks are employed to capture long-term dependencies in Spatio-temporal data, while attention mechanisms help to focus on relevant input data during training, thereby improving prediction accuracy and reducing computational resource consumption. The attention model can learn to weigh certain input features more heavily than others, based on their relevance to the prediction task. By considering the pertinent information, attention mechanisms are used to reduce the amount of computational resources consumption and faster training time. The ability to capture long-term dependencies and focus on relevant input data leads to more accurate predictions of air quality. Figure 5 shows the block diagram of Attention-based Long Short-Term Memory for the air quality prediction model.

Attention-based Long short-term memory for air quality prediction model diagram is shown in the above Fig. 5. The first gate in Attention-based Long Short-Term Memory for air quality prediction is the forget gate ‘ for_t ’. The forget gate determines the air quality information to be eliminated in the memory cells. The forgotten data determines which information in the memory cell ‘ C_{t-1} ’ from the previous instance to be passed to memory cell ‘ C_t ’ at the present instance. The forget gate is formulated as given below.

$$for_t = \text{sigma}(W_f \cdot [h_{t-1}, FS_t] + b_f) = \text{sigma}(W_{hf}h_{t-1} + W_{FSf} + b_f) \quad (16)$$

From the above Eq. (16), ‘ sigma ’, represents the sigmoid function mapping variables between ‘0’ and ‘1’ and ‘ h_{t-1} ’ denotes the previous hidden vector. Moreover, ‘ for_t ’ refers to the forget gate at time instance ‘ t ’; ‘ h_t ’ refers to the hidden vector with ‘ W_{hf} , ‘ W_{FSf} ’ representing the weight matrices (i.e., hidden-forget and input-forget matrices) respectively, the ‘ b_f ’ represents the bias vectors in the forget gate, for an input sequence ‘ $FS = \{FS_1, FS_2, \dots, FS_n\}$ ’ respectively. The second gate forms the input gate ‘ Inp_t ’ that decides how much new input air quality information enters the memory cell ‘ C_t ’ at the present instance. C_t at the current moment. The input gate is formulated as given below.

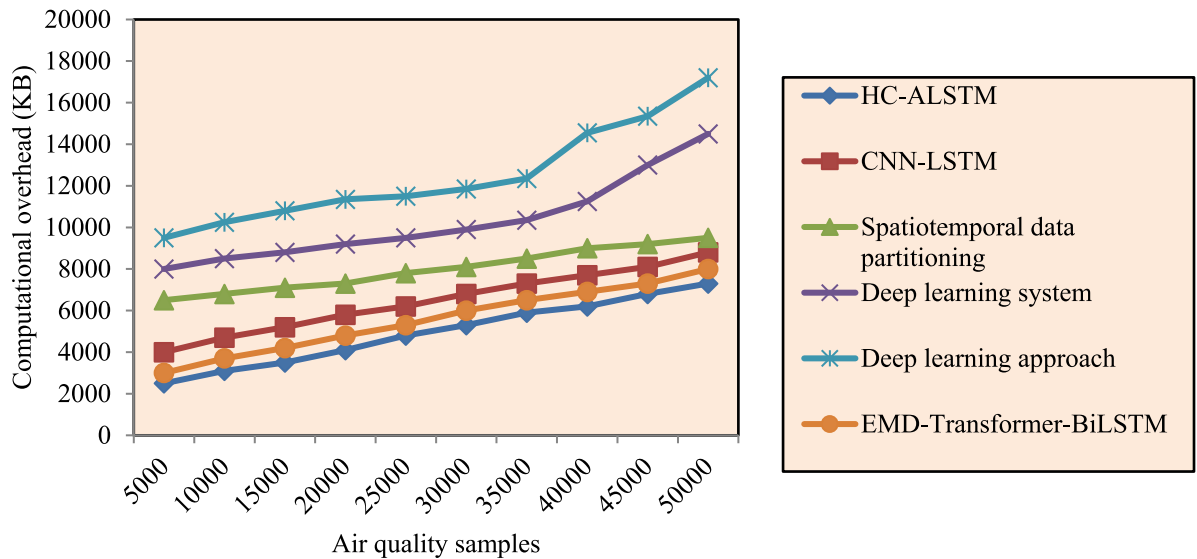


Fig. 6. Comparative analysis of computational overhead using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM.

$$Inp_t = \text{sigma}(W_{Inp} \cdot [h_{t-1}, FS_t] + b_{Inp}) = \text{sigma}(W_{hInp} h_{t-1} + W_{FSFS_t} + b_{Inp}) \quad (17)$$

From the above Eq. (17), ' W_{Inp} ' denotes the input gate weight at time instance ' t ' and ' b_{Inp} ' denotes the input gate bias vectors respectively, in the input gate, with ' W_{hInp} , W_{FSFS_t} ' representing the weight matrices (i.e., hidden-input and feature-selected matrices). The third gate is the output gate that determines whether the air quality information in the memory cell ' C_t ' at present instance can get into the hidden layer ' h_t '. The output gate is formulated as given below.

$$Out_t[AQI] = \text{sigma}(W_{Out} \cdot [h_{t-1}, FS_t] + b_{Out}) = \text{sigma}(W_{hOut} h_{t-1} + W_{FSOut} + b_{Out}) \quad (18)$$

From the above Eq. (18), ' Out_t ' refers to the output gate at the time ' t ' with the corresponding output bias vector denoted by ' b_{Out} ' respectively. Finally, the Attention-based Long Short-Term Memory for air quality prediction requires updating the memory cell. Memory cell updates are made using the forget and input gates, which use the context vector to remember the long input source.

The attention mechanism allows the model to capture the most vital features of past states. In order to take advantage of the information of the past states, an attention-based layer is added to the LSTM layer in the proposed HC-ALSTM method. An attention layer is employed to allocate weights to sample data and extract vital information that can impact the features analysis and forecasting process. This process offers a better estimate of the final output and minimizes the resource consumption of the computer. The attention layer generates the context vector, a weighted combination of input states to remember the significant sequence while training. Therefore, the context vector is employed to process the entire input sequence, thus eliminating the problem of forgetting. The context vector ' C_t ' is the aggregate sum of hidden states ' $H = (h_1, h_2, \dots, h_{t-1})$ ' for the current step mathematically formulated below.

$$C_t = \sum_{i=1}^{t-1} \alpha_i H_i, \text{ where } H = (h_1, h_2, \dots, h_{t-1}) \quad (19)$$

$$\alpha_i = \frac{\exp[f(h_i, h_t)]}{\sum_{j=1}^{t-1} \exp[f(h_j, h_t)]} \quad (20)$$

$$h_t = Out_t[AQI] * \tanh(C_t) \quad (21)$$

From the above Eqs. (19), (20), and (21) results will output the information. The data is processed by ' $\tanh(C_t)$ '. The AQI values and corresponding air quality classification are listed in Table 4.

With the aid of the above AQI value and the air quality level, the results obtained from the output unit in the output layer and processed by ' $\tanh(C_t)$ ' are measured, and accordingly, the prediction results are provided. The pseudo-code representation of the Attention-based Long Short-Term Memory for air quality prediction is given below.

Input: Dataset ‘ DS ’, Features ‘ $F = \{F_1, F_2, \dots, F_m\}$ ’, Station Features ‘ $SF = \{SF_1, SF_2, \dots, SF_{sm}\}$ ’, Samples ‘ $S = \{S_1, S_2, \dots, S_n\}$ ’
Output: Resource-efficient, accurate air pollution prediction
1: Initialize features selected ‘ FS ’ 2: Begin 3: For each Dataset ‘ DS ’ with Features ‘ F ’, Station Features ‘ SF ’, features extracted ‘ FE ’, features selected ‘ FS ’ and processed sample results ‘ PS ’ 4: Formulate forget gate ‘ for_t ’ as given in equation (16) 5: Formulate the input gate ‘ Inp_t ’ as given in equation (17) 6: Formulate the output gate ‘ $Out_t[AQI]$ ’ as given in equation (18) 7: Obtain cell state based on the context vector output ‘ C_t ’, ‘ α_t ’ and ‘ h_t ’ as given in equations (19), (20), and (21) 8: If ‘ AQI lies between 0 & 50’ 9: Then ‘Air Quality Level = Excellent’ 10: End if 11: If ‘ AQI lies between 51 & 100’ 12: Then ‘Air Quality Level = Good’ 13: End if 14: If ‘ AQI lies between 101 & 150’ 15: Then ‘Air Quality Level = Light pollution’ 16: End if 17: If ‘ AQI lies between 151 & 200’ 18: Then ‘Air Quality Level = Moderate pollution’ 19: End if 20: If ‘ AQI lies between 201 & 300’ 21: Then ‘Air Quality Level = Heavy pollution’ 22: End if 23: If ‘ $AQI > 301$ ’ 24: Then ‘Air Quality Level = Serious pollution’ 25: End if 26: Return ‘Air Quality Level’ results 27: End for 28: End

Algorithm 4. Attention-based Long Short-Term Memory for Air Quality Prediction.

Step-by-step process of Attention-based Long Short-Term Memory for Air Quality Prediction is explained in Algorithm 4. The attention mechanism is applied as a context vector in the output layer to improve the accuracy in a resource-efficient manner even in case of a long-term dependence on spatiotemporal. First, the three gates, i.e., forget gate, input gate, and output gate, are formulated separately for the extracted and selected features. Second, cell state, i.e., the air quality data to be retained and those that should be discarded, are then performed using context vector output. With the attention mechanism, the context vector holds all the input sequences, eliminating the forgetting issue by remembering the significant sequence. This, in turn, not only improves the accuracy rate of air quality prediction but also reduces the overhead involved in the entire process.

Experimental settings

In this section, the performance air quality prediction called HC-ALSTM is performed using Python. To measure the HC-ALSTM method, Two datasets namely Air Quality Data in India and Global Air Pollution Dataset

Air quality samples	Air quality prediction time (ms)					
	HC-ALSTM	CNN-LSTM	Spatiotemporal data partitioning	Deep learning system	Deep learning approach	EMD-Transformer-BiLSTM
5000	1100	1400	1550	1750	2100	1205
10,000	1135	1535	1735	2000	2535	1285
15,000	1155	1815	2025	2235	2735	1465
20,000	1185	2055	2155	2545	3035	1575
25,000	1255	2135	2355	2835	3135	1725
30,000	1315	2255	2595	3155	3855	1745
35,000	1455	2835	3135	3835	4155	2115
40,000	1575	3135	3555	4255	4535	2365
45,000	1835	3355	4015	4855	5055	2755
50,000	2025	3825	4215	5325	5385	3115

Table 8. Air quality prediction time estimates using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM.

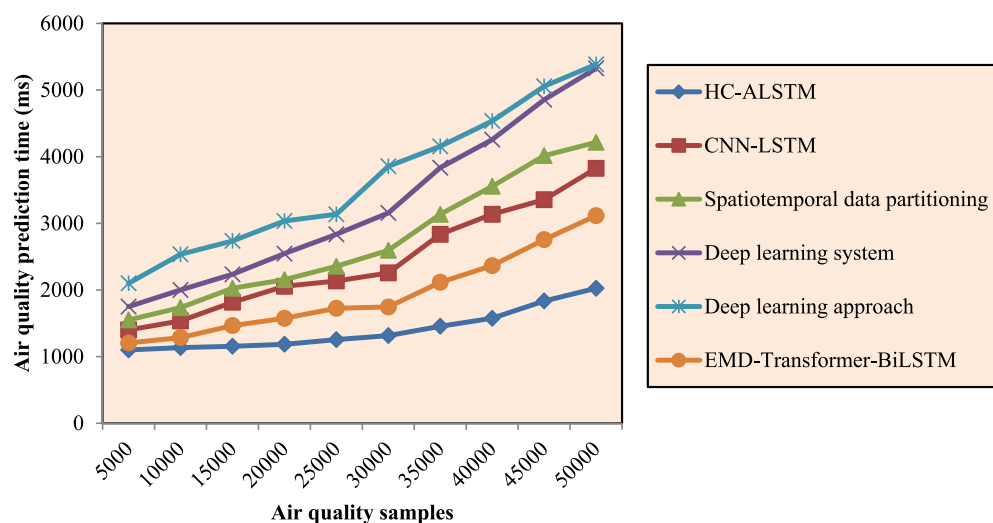


Fig. 7. Comparative analysis of air quality prediction time using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM.

employed. Air Quality Data in India obtained from <https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india> is used for analysis. Global Air Pollution Dataset taken from the <https://www.kaggle.com/datasets/sibalmuzdadid/global-air-pollution-dataset>. Experiments are conducted on factors such as air quality prediction time, air quality prediction accuracy, root mean square error, and overhead to distinct air quality samples. A fair comparison is made with the three existing methods, CNN-LSTM¹, Spatiotemporal data partitioning², and EMD-Transformer-BiLSTM²⁹ and two state-of-the-art methods, deep learning system³ and deep learning approach⁴ for 10 simulation runs. The hardware and software configurations are illustrated in Table 5.

Table 6 given below lists the hyper-parameters and their description utilized in our proposed method.

Implementation details

In this study, we developed a proposed HC-ALSTM method for air quality prediction.

- The HC-ALSTM method includes preprocessing, feature extraction, feature selection, and prediction.
- We compared our HC-ALSTM method and existing CNN-LSTM¹, Spatiotemporal data partitioning², EMD-Transformer-BiLSTM²⁹, and two state-of-the-art methods, deep learning system³ and deep learning approach⁴ using two different datasets to validating the results.
- Initially, raw air quality data was taken as input. The Statistical Normalization-based Preprocessing is employed to normalize the preprocessed data.
- Spatio-temporal features are extracted with Generalised Hebbian Spatio-temporal Feature extraction.
- Concordance Correlation is applied to chosen pertinent features and eradicates irrelevant features.
- Attention-based Long Short-Term Memory employed three gates namely forget gate, input gate, and output gate for air quality prediction. Also, the context vector estimate achieves an aggregate sum of hidden states for classifying the air quality levels.

Air quality samples	Air quality prediction accuracy (%)					
	HC-ALSTM	CNN-LSTM	Spatiotemporal data partitioning	Deep learning system	Deep learning approach	EMD-Transformer-BiLSTM
500	99.1	98.5	98.1	96.5	95.1	94.95
1000	97.35	96	95.45	94.15	93.25	91.45
1500	97	95.55	94	93.55	91	89
2000	96.85	95.35	93.25	92	90.55	88.65
2500	96.55	94	93	91.55	89.25	88
3000	96	93.25	91.55	89	86	84.45
3500	95.45	93	91	86.35	84.25	82.15
4000	94.15	91.55	89.25	86	83.15	81.45
4500	94	91.25	88	85.35	83	80
5000	93.85	91	87.35	84	83	79

Table 9. Air quality prediction accuracy estimates using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM

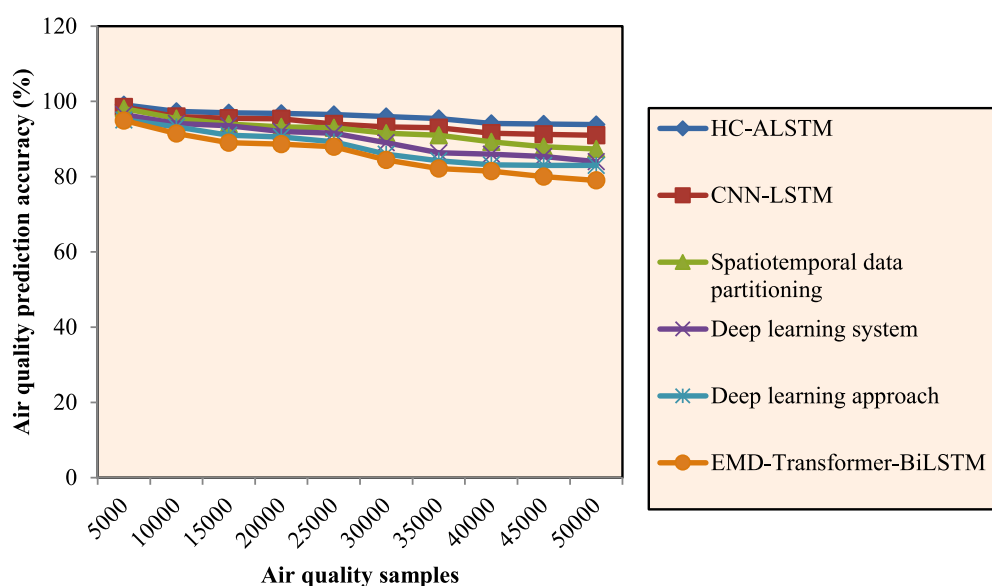


Fig. 8. Comparative analysis of air quality prediction accuracy using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM.

Based on the above implementation patterns, four different evaluation metrics are detailed in the next section.

Discussion

Performance analysis of computational overhead

The first performance metric for analysis is computational overhead. The computational overhead in air pollution prediction involves predicting the atmosphere's composition for a given period and location. Also, a proportion of resources are said to be consumed during this process. The computational overhead involved is mathematically represented below.

$$CO = \sum_{i=1}^n S_i * Mem [AQI] \quad (22)$$

From the above Eq. (22), the computational overhead 'CO' is measured based on the air quality samples involved in the simulation process ' S_i ' and the memory involved in air quality prediction ' $Mem [AQI]$ '. It is measured in terms of kilobytes (KB). Table 7 given below, lists the results of modeling the computational overhead.

Figure 6 above shows the computational overhead with respect to air quality samples. In the figure, the x-axis represents the air quality samples range from 5000 to 50,000 collected from air quality data in the India dataset from 2015 to 2020. y-axis indicates the computational overhead for six methods namely proposed HC-ALSTM and existing CNN-LSTM, spatiotemporal data partitioning, deep learning system, and deep learning approach.

Air quality samples	RMSE					
	HC-ALSTM	CNN-LSTM	Spatiotemporal data partitioning	Deep learning system	Deep learning approach	EMD-Transformer-BiLSTM
5000	0.14	0.19	0.24	0.28	0.35	0.17
10,000	0.17	0.22	0.27	0.33	0.41	0.19
15,000	0.19	0.25	0.31	0.38	0.45	0.23
20,000	0.22	0.29	0.35	0.44	0.52	0.26
25,000	0.24	0.32	0.4	0.49	0.57	0.28
30,000	0.27	0.37	0.42	0.52	0.6	0.31
35,000	0.28	0.4	0.48	0.55	0.65	0.35
40,000	0.31	0.44	0.55	0.63	0.71	0.39
45,000	0.35	0.48	0.59	0.65	0.78	0.42
50,000	0.4	0.55	0.62	0.72	0.82	0.48

Table 10. RMSE estimates using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM.

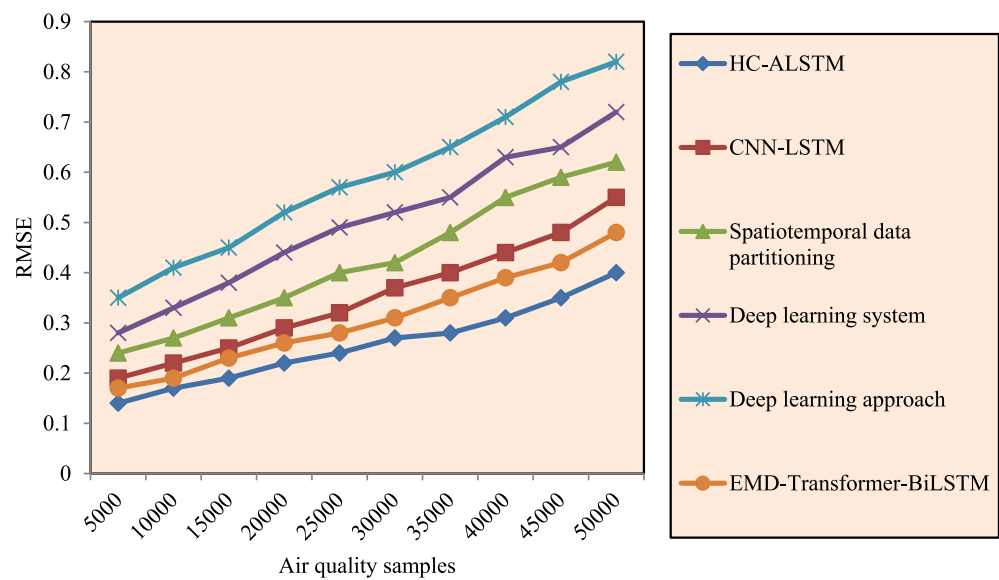


Fig. 9. Comparative analysis of RMSE using HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach and EMD-Transformer-BiLSTM.

From the above figure, for different years or with the increase in the air quality of different cities and the increase in the population in a parallel fashion, it causes a depletion of planetary health equally. Here, resources are said to be utilized while ensuring the safe concentration of pollutants in the specific city. However, the computational overhead using the proposed HC-ALSTM method for 2015–2020 was found to be comparatively lesser than the three traditional methods, CNN-LSTM¹, Spatiotemporal data partitioning², EMD-Transformer-BiLSTM²⁹ and two state-of-the-art methods, Deep learning system³, Deep learning approach⁴ respectively. The reason behind the minimization of computational overhead, even in the case of increasing population or big data, was the integration of the Concordance Correlation-based Feature selection algorithm. By applying this algorithm, the dynamic correlation of both ‘PM2.5’ and ‘PM10’ considering both the previous and future time points were captured using the Concordance Correlation function. Also, using variance and covariance measures, not only relevant features were selected, but the storage or relevant features in the stack also resulted in the minimization of computational overhead. With this, the computational overhead using the proposed HC-ALSTM method was said to be reduced by 25% compared to¹, 39% compared to², 53% compared to³, and 61% compared to⁴, and 12% compared to²⁹ respectively.

Performance analysis of air quality prediction time

One of the world’s most significant issues is air pollution, which causes respiratory issues, lung diseases, and cardiovascular problems, therefore contributing to not only mental health issues but also, in some instances, aggravating prevailing health conditions. Early air quality prediction in a particular location can assist the general public in aggravating health conditions. Therefore, air quality prediction time remains one of the second

performance metrics used in the analysis. The air quality prediction time is mathematically formulated as given below.

$$AQP_{time} = \sum_{i=1}^n Sample_i * Time [AQP] \tag{23}$$

From the above Eq. (23), the air quality prediction time ‘ AQP_{time} ’ is estimated based on the samples involved in air quality prediction ‘ $Sample_i$ ’ and the time consumed in air quality prediction ‘ $Time [AQP]$ ’ respectively. It is measured in milliseconds (ms). The results of air quality prediction time are listed in Table 8.

Figure 7 given above shows the graphical representation of air quality prediction time. With the varying air quality samples obtained from 231 different cities between 2015 and 2020, they increased the air quality samples, resulting in an increase in the number of features to be analyzed and an increase in the air quality prediction time. However, with 5000 samples involved in the simulation process, the prediction time using the five methods was observed to be 1100 ms, 1400 ms, 1550 ms, 1750 ms, 2100 ms, and 1205 ms respectively. From this inference, the air quality prediction time using the HC-ALSTM method is said to be comparatively lesser than the two traditional methods,^{1,2,29} and two state-of-the-art methods,^{3,4} respectively. The improvement was due to applying the Generalised Hebbian Spatio Temporal Feature extraction algorithm. Moreover, spatial factors were obtained by employing the aggregated vector representation. With this, only significant and relevant features involved in further processing were utilized that, in turn, reduced the air quality prediction time using the HC-ALSTM method by 40% compared to¹, 46% compared to², 55% compared to³, 60% compared to⁴ and 25% compared to²⁹ respectively.

Performance analysis of air quality prediction accuracy

While predicting the air quality of a city or station, with big data analysis involved, accuracy remains one of the major factors influencing the overall performance of the method. Therefore, air quality prediction accuracy remains one of the major performance metrics in big data. The accuracy of air quality prediction is mathematically formulated, as given below.

$$AQP_{acc} = \sum_{i=1}^n \frac{Sample_{ap}}{Sample_i} * 100 \tag{24}$$

From the above Eq. (24), air quality prediction accuracy ‘ AQP_{acc} ’ is measured based on the samples involved in analyzing the air quality ‘ $Sample_i$ ’ and the samples accurately predicted ‘ $Sample_{ap}$ ’. The higher the prediction accuracy, the better the method is said to be. It is measured in terms of percentage (%). Table 9 summarizes the results of modelling the prediction accuracy.

Figure 8 given above presents a comparative air quality prediction accuracy performance. Specifically, the accuracy of the proposed HC-ALSTM method is 99.1% for predicting the air quality of five different cities, i.e., Chennai, Coimbatore, Delhi, Kochi, and Talcher, in the years between 2015 and 2018. In contrast, those for the two traditional methods, CNN-LSTM¹ and spatiotemporal data partitioning² and EMD-Transformer-BiLSTM²⁹ were found to be 95.5%, 98.1%, 94.95%, and two state-of-the-art methods, deep learning system³ and deep learning approach⁴ being 96.5%, 95.1% respectively. As shown in Fig. 7 above, the performance of the HC-ALSTM method is more efficient than that of the other three traditional methods and two state-of-the-art methods for measuring air quality. This was owing to applying the Generalised Hebbian Spatio Temporal Feature extraction algorithm. Three different vector representations are formed separately based on temporal factors. Next, with these three vector representations, an aggregated vector was obtained, and the Generalised Hebbian function was applied, which also involved spatial factors for further processing. This, in turn, resulted in the improvement of air quality prediction accuracy using the HC-ALSTM method by 2%, 4% and 12% compared to the traditional methods, CNN-LSTM¹, spatiotemporal data partitioning², EMD-Transformer-BiLSTM²⁹ and improved by 7% and 9% compared to two state-of-the-art methods, deep learning system³ and deep learning approach⁴ respectively.

Performance analysis of RMSE

In order to evaluate the efficiency of the air quality prediction method proposed in this paper, an evaluation index, Root Mean Square Error (RMSE), was adopted to estimate the error between prediction results and true

Methods	HC-ALSTM	CNN-LSTM	Spatiotemporal data partitioning	Deep learning system	Deep learning approach	EMD-Transformer-BiLSTM
Computational overhead (KB)	4700	6270	7650	10,100	12,260	5200
Air quality prediction time (ms)	1253	2272	2545	3026	3475	1715
Air quality prediction accuracy (%)	94.85	90.26	88.55	87.58	85.29	83.75
RMSE	0.21	0.33	0.39	0.44	0.52	0.27

Table 11. performance analysis results of HC-ALSTM, CNN-LSTM, spatiotemporal data partitioning, deep learning system, deep learning approach, and EMD-Transformer-BiLSTM using Global Air Pollution Dataset.

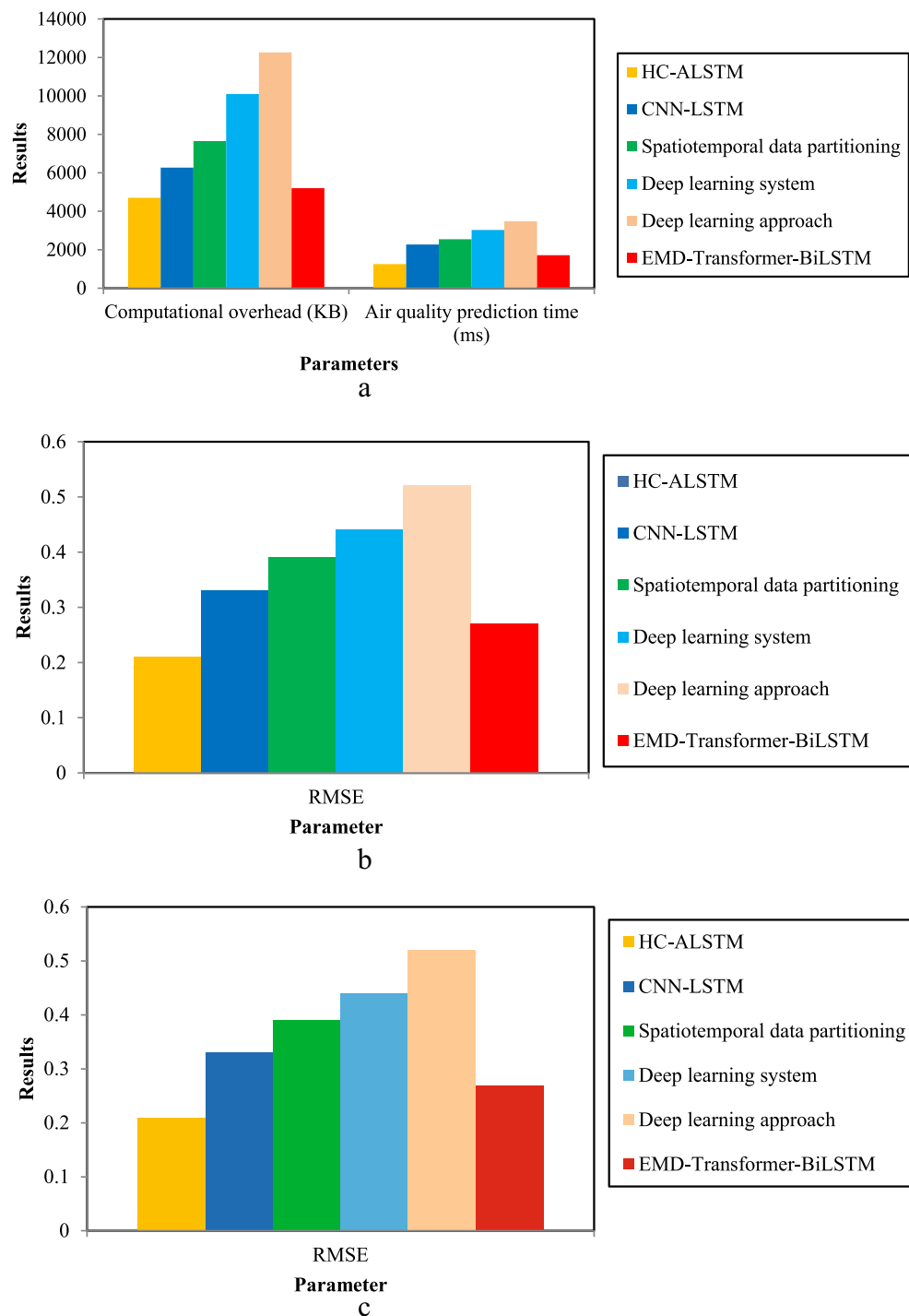


Fig. 10. (a) Overall performance results of computation overhead, air quality prediction time. (b) Overall performance results of air quality prediction accuracy. (c) Overall performance results of RMSE.

values. To be more specific, the smaller the RMSE values are, the smaller the divergence between predicted results and true values is, and the better the proposed method is. The calculation formula of RMSE is given below.

$$RMSE(S, S') = \sqrt{\frac{1}{n} \sum_{i=1}^n (S_i - S'_i)^2} \quad (25)$$

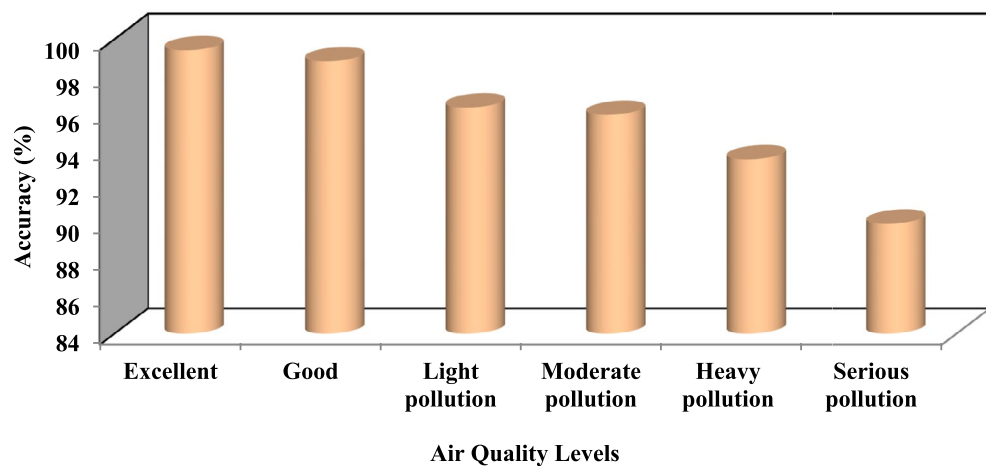


Fig. 11. Ablation experiment results graph for classification.

From the above Eq. (25), the root mean square error ' $RMSE$ ' value is obtained based on the real sample data ' S_i ' and the prediction results of the proposed method ' $\hat{S}_i$ ' concerning the sum of ' n ' data samples. The lower the prediction error, the more efficient the air quality data analysis is said to be made. Table 10 given below summarizes the results of modeling the RMSE.

Finally, Fig. 9 given above illustrates the RMSE values of the city and station-India's air quality data from 2015 to 2020 predicted by the proposed method, HC-ALSTM, three traditional methods, CNN-LSTM¹, spatiotemporal data partitioning², EMD-Transformer-BiLSTM²⁹ and two state-of-the-art methods, deep learning system³ and deep learning approach⁴ respectively. It can be inferred from the above figure that the model proposed in this paper (Statistical Normalization-based Preprocessing) is significantly superior to other comparison methods. Among them, in terms of RMSE, the HC-ALSTM method proposed in this paper has achieved a maximum improvement of 30%, 44%, and 16% using^{1,2,29}, 51%, 56% using^{3,4} respectively, compared with the classical air quality data analysis methods, CNN-LSTM¹, spatiotemporal data partitioning², deep learning system³, deep learning approach⁴ and EMD-Transformer-BiLSTM²⁹. These results illustrate the multivariate input proposed in this paper. These results demonstrate the efficiency and advancement of the multivariate input operation via four input data matrices. At the sometime, by comparing the two traditional methods, CNN-LSTM¹, spatiotemporal data partitioning², and EMD-Transformer-BiLSTM²⁹, proposed in this paper, the RMSE was improved by 26%, 39%, and 16%. Similarly, by comparing the two state-of-the-art methods, a deep learning system³ and a deep learning approach⁴ were improved by 49% and 57%. By significantly minimizing the error rate, the proposed method can provide a means for time and early prediction.

Comparison of proposed and existing methods using the global air pollution dataset

In this section, performance analysis of the proposed HC-ALSTM method, CNN-LSTM¹, Spatiotemporal data partitioning², deep learning system³, deep learning approach⁴, and EMD-Transformer-BiLSTM²⁹ are compared with Global Air Pollution Dataset. The evaluation is conducted using various metrics, such as computational overhead, air quality prediction time, accuracy of air quality prediction, and RMSE. Table 11 given below summarizes the results of HC-ALSTM,^{1-4,29} using the Global Air Pollution Dataset.

Figure 10a–c demonstrate the overall performance results of all parameters, including computational overhead, air quality prediction time, the accuracy of air quality prediction, and RMSE, for six methods namely HC-ALSTM,^{1-4,29}. These results clearly indicate that the performance of all parameters using the proposed HC-ALSTM method is significantly improved compared to the existing methods.

Ablation study

Ablation study is the procedure of eliminating sections of input systematically which elements are inputs similar to output. In our work, we have used preprocessing for preprocessing the output data. Next, feature extraction and selection are employed to extract the Spatio-temporal features and pick relevant features. Attention-based LSTM classifies different air quality levels namely Excellent, Good, Light pollution, Moderate pollution, Heavy pollution, and Serious pollution.

Figure 11 shows the ablation experiment results graph for classification. In the above figure, the x-axis refers to the various air quality levels and the y-axis gives the accuracy in percentage (%) for the proposed HC-ALSTM method.

Conclusion

Deep learning techniques act as a purging solution to handle air quality prediction via air quality big data analysis that is swiftly heightening the consciousness of researchers and academics. The research mentioned above identifies that there is a requirement to design a notable method to ensure accurate prediction and control measures to be taken accordingly based on air quality data. To be more definite, addressing the accuracy, time,

and error factors involving air quality data analysis prediction is essential to ease planning day-to-day chores, circumventing locations with high alerts, and implementing efficient pollution control measures for the general public. Hence, this work aims to propose air quality data analysis via deep learning technique to develop a Hebbian Concordance and Attention-based Long Short-Term Memory (HC-ALSTM) method that ensures prediction in an accurate and timely manner with minimum error. The entire method is designed under the deep neural learning framework. First, a Statistical Normalization-based Preprocessing model was designed to normalize raw data. Second, Generalised Hebbian Spatio Temporal Feature extraction was designed to model error-minimized optimal air quality prediction. Third, to focus on the spatial and temporal data, the Generalised Hebbian Spatio Temporal Feature extraction algorithm was designed with which accurate predictions are said to be made. Finally, an error was minimized, and accurate prediction was ensured using Attention-based Long Short-Term Memory. The experimental results show that the HC-ALSTM method can get better results regarding air quality prediction accuracy, air quality prediction time, error, and computational overhead using Air Quality data in the India dataset, which fully shows that applying it to deep paves the way for robust decision making.

Key findings

From the experimental results, the following summary key findings are achieved as follows,

- The proposed HC-ALSTM method achieved higher air quality prediction accuracy by 7% when compared HC-ALSTM method, CNN-LSTM¹, Spatiotemporal data partitioning², deep learning system³, deep learning approach⁴ and EMD-Transformer-BiLSTM²⁹.
- Computational overhead is reduced up to 38% using the proposed HC-ALSTM method when compared to existing methods.
- Also, the proposed HC-ALSTM method of air quality prediction time is minimized by 45% than the conventional methods.
- RMSE is decreased up to 37% with the proposed HC-ALSTM method compared to the HC-ALSTM method,^{1–4,29}.

Limitations and future work

The limitation of the HC-ALSTM method is failed to handle the missing value and outlier data removal. In addition, they failed to focus on various parameters such as precision and recall for achieving air quality prediction performance. In the future, different preprocessing methods will be performed to fill in missing data and detect the outliers. Also, a novel deep learning method is recommended for classifying air quality levels by considering precision and recall.

Data availability

Data are publicly available within the article. The data have been gathered from <https://www.kaggle.com/dataset/s/rohanrao/air-quality-data-in-india>. Hourly data of the air quality indices were obtained from the cities such as Ahmedabad, Aizawl, Amaravati, Amritsar, Bengaluru, Bhopal, Brajrajnagar, Chandigarh, Chennai, Coimbatore, Delhi, Ernakulam, Gurugram, Guwahati, Hyderabad, Jaipur, Jorapokhar, Kochi, Kolkata, Lucknow, Mumbai, Patna, Shillong, Talcher, Thiruvananthapuram, Visakhapatnam on 2015–2020.

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Author contributions

All the authors have participated in writing the manuscript and have revised the final version. All authors read and approved the final manuscript.

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Declarations

Competing interests

The authors declare no conflicts of interest to report regarding the present study.

Ethical approval

No human or animal studies were conducted by any of the authors for this article.

Additional information

Correspondence and requests for materials should be addressed to S.S.

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